

T0-Theory: Cosmic Relations

Contents

.1	Introduction: The Universal ξ -Constant	2
	Foundations of T0 Theory	2
	Time-Energy Duality as Foundation	2
.2	Cosmic Microwave Background (CMB)	3
	CMB without Big Bang: ξ -Field Mechanisms	3
	CMB Energy Density and ξ -Length Scale	3
.3	Casimir Effect and ξ -Field Connection	4
	Casimir-CMB Ratio as Consistency Check	4
	ξ -Field as Universal Vacuum	4
.4	Cosmic Redshift without Expansion	5
	ξ -Field Energy Loss Mechanism	5
	Wavelength-Dependent Redshift	5
.5	Structure Formation in the Static ξ -Universe	6
	Continuous Structure Development	6
	ξ -Supported Continuous Creation	6
.6	Dimensionless ξ -Hierarchy	6
	Energy Scale Ratios	6
.7	Experimental Predictions and Tests	7
	Precision Measurements of Gravitational Constant	7
	Casimir Force Anomalies	7
	Electromagnetic Resonance	7
.8	Cosmological Consequences	7
	Solution to Cosmological Problems	7
	Parameter Reduction	8

Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- CMB temperature: $\frac{16}{9}\xi^2 \cdot 7500$: $2.35 \times 10^{-4} \rightarrow 2.37 \times 10^{-4}$; in SI $2.725 \text{ K} \rightarrow 2.751 \text{ K}$ (see R70).
- $\rho_{\text{CMB}} = 4.87 \times 10^{41}$: cannot be reproduced, annotated.
- Casimir: energy density $\pi^2/(240 d^4) \rightarrow \pi^2/(720 d^4)$ (240 is the pressure); ratio to ρ_{CMB} $308 \rightarrow 102.8$, direct calculation $312 \rightarrow 104$ (calculated, not a measured value), also in the equation and table further below.
- Resonance frequency: $\nu_\xi = 1/L_\xi = 10^4 \text{ Hz} \rightarrow c/L_\xi \approx 3 \times 10^{12} \text{ Hz}$.
- G_{T0} : already corrected in the text on 29 Sept. 2026.
- G precision (note of 1 Oct. 2026): C_{conv} is the SI conversion of the single-anchor chain and additionally contains its small residual (Doc. 383); the agreement to 0.003 % does not count as an independent precision result (Doc. 190, R141).
- (2 Oct. 2026) Notation f : function $f(E/E_\xi)$ — not a frequency; note in the text (Doc. 190, R147).

Abstract

The T0-theory examines how far cosmic phenomena can be traced back to a single universal constant $\xi = \frac{4}{3} \times 10^{-4}$. This document presents the fundamental relationships between the gravitational constant, cosmic microwave background radiation (CMB), Casimir effect and cosmic structures within the framework of a static, eternally existing universe. All derivations are performed in natural units ($\hbar = c = k_B = 1$) and respect the time-energy duality as a fundamental principle of quantum mechanics.

Note on versions. The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus. This document deals almost exclusively with the cosmic sector (static universe, CMB, redshift, dark energy). The current corpus deliberately leaves this sector aside, because the Standard Model does not derive it either (P39); the cosmological conclusions of this document are therefore to be read as **[S]**.

.1 Introduction: The Universal ξ -Constant**Foundations of T0 Theory****Important**

T0 theory is based on the universal dimensionless constant $\xi = \frac{4}{3} \times 10^{-4}$, to which it traces phenomena from the subatomic to the cosmic scale.

T0 theory is built on a single fundamental constant. This constant forms the basis for all physical calculations and predictions of the theory:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333333... \times 10^{-4} \quad (1)$$

This dimensionless constant connects quantum and gravitational phenomena and is intended as the basis of a unified description of the fundamental interactions.

Note on Derivation

For the detailed derivation and physical justification of this fundamental constant, see the document "Parameter Derivation" (available at:).

Time-Energy Duality as Foundation

Thesis

[S] From Heisenberg's uncertainty relation $\Delta E \times \Delta t \geq \hbar/2 = 1/2$ (natural units) this document concludes that a classical Big Bang singularity with infinite density is not physically realized and is replaced by a tiny but finite core with minimal length scale L_0 (from ξ).

Heisenberg's uncertainty relation between energy and time represents the fundamental principle of T0-theory:

$$\Delta E \times \Delta t \geq \frac{1}{2} \quad (\text{natural units}) \quad (2)$$

In the reading of this document, this relation has far-reaching cosmological consequences **[S]**:

- A temporal beginning (Big Bang) would mean $\Delta t = \text{finite}$
- This leads to $\Delta E \rightarrow \infty$ - physically inconsistent
- Therefore the universe must have existed eternally: $\Delta t = \infty$
- The universe is static, without expanding space

.2 Cosmic Microwave Background (CMB)

CMB without Big Bang: ξ -Field Mechanisms

Thesis

[S] If time-energy duality excludes a Big Bang, as concluded above, the CMB must have a different origin than the $z=1100$ decoupling of standard cosmology.

T0-theory interprets the CMB **[S]** as radiation from ξ -field quantum fluctuations:

$$\frac{T_{\text{CMB}}}{E_{\xi}} = \frac{16}{9} \xi^2 \quad (3)$$

With $E_{\xi} = \frac{1}{\xi} = \frac{3}{4} \times 10^4$ (natural units) and $\xi = \frac{4}{3} \times 10^{-4}$ this yields:

$$T_{\text{CMB}} = \frac{16}{9} \xi^2 \times E_{\xi} = \frac{16}{9} \times 1.78 \times 10^{-8} \times 7500 = 2.37 \times 10^{-4} \quad (4)$$

(Corrected on 30 Sept. 2026: previously 2.35×10^{-4} – with $\xi = \frac{4}{3} \times 10^{-4}$, $\frac{16}{9}\xi^2 \cdot 7500$ gives 2.370×10^{-4} .)

Conversion to SI units:

$$T_{\text{CMB}} = 2.751 \text{ K} \quad (5)$$

(Corrected on 30 Sept. 2026: previously 2.725 K – that is the measured value; 2.370×10^{-4} (with E_ξ in eV) corresponds to 2.751 K.)

The expression thus lies about 1 % from the observed value (cf. the table of dimensionless ratios: 3.13 versus 3.16×10^{-8}).

CMB Energy Density and ξ -Length Scale

The CMB energy density in natural units is:

$$\rho_{\text{CMB}} = 4.87 \times 10^{41} \quad (\text{natural units, dimension } [E^4]) \quad (6)$$

(Note of 30 Sept. 2026: the value 4.87×10^{41} cannot be reproduced; in natural units with energy in eV, $\rho_{\text{CMB}} = \frac{\pi^2}{15} T_{\text{CMB}}^4$ gives about $2.0 \times 10^{-15} \text{ eV}^4$.)

This energy density defines a characteristic ξ -length scale:

$$L_\xi = \left(\frac{\xi}{\rho_{\text{CMB}}} \right)^{1/4} \quad (7)$$

Fundamental relation of CMB energy density:

$$\rho_{\text{CMB}} = \frac{\xi}{L_\xi^4} = \frac{\frac{4}{3} \times 10^{-4}}{(L_\xi)^4} \quad (8)$$

.3 Casimir Effect and ξ -Field Connection

Casimir-CMB Ratio as Consistency Check

Experiment

The ratio between Casimir energy density and CMB energy density is consistent with the characteristic ξ -length scale of $L_\xi = 10^{-4} \text{ m}$.

The Casimir energy density at plate separation $d = L_\xi$ is:

$$|\rho_{\text{Casimir}}| = \frac{\pi^2}{720 \times L_\xi^4} \quad (\text{natural units}) \quad (9)$$

(Corrected on 30 Sept. 2026: previously denominator 240 – according to the literature, $\pi^2 \hbar c / (240 d^4)$ is the Casimir pressure between ideal plates; the energy density is $\pi^2 \hbar c / (720 d^4)$.)

The ratio yields:

$$\frac{|\rho_{\text{Casimir}}|}{\rho_{\text{CMB}}} = \frac{\pi^2}{720 \xi} = \frac{\pi^2 \times 10^4}{960} \approx 102.8 \quad (10)$$

(Corrected on 30 Sept. 2026: previously $\pi^2 / (240 \xi) = \pi^2 \times 10^4 / 320 \approx 308$, labelled as the “experimental” ratio – with the energy density (denominator 720) the ratio is 102.8; it is calculated, not measured.)

Direct calculation with SI values: With $L_\xi = 10^{-4}$ m, direct calculation gives:

$$|\rho_{\text{Casimir}}| = \frac{\hbar c \pi^2}{720 \times (10^{-4})^4} = 4.33 \times 10^{-12} \text{ J/m}^3 \quad (11)$$

$$\rho_{\text{CMB}} = 4.17 \times 10^{-14} \text{ J/m}^3 \quad (12)$$

$$\text{Ratio} = \frac{4.33 \times 10^{-12}}{4.17 \times 10^{-14}} = 104 \quad (13)$$

(Corrected on 30 Sept. 2026: previously denominator 240, 1.3×10^{-11} J/m³ and ratio 312 – 1.3×10^{-11} is the Casimir pressure in Pa, the energy density is one third of it; the ratio is calculated, not a measured value, only ρ_{CMB} is measured.)

The theoretical expression (102.8) and the direct calculation with the measured ρ_{CMB} (104) differ by about 1%. (Corrected on 30 Sept. 2026: previously 308, 312 and 1.3% – follow-up values of the correction to the Casimir energy density.)

ξ -Field as Universal Vacuum

Important

The ξ -field manifests both in free CMB radiation and in geometrically constrained Casimir vacuum. In the interpretation of this document this is a hint at the physical reality of the ξ -field **[S]**; it is not a proof.

The characteristic ξ -length scale L_ξ is the point where CMB vacuum energy density and Casimir energy density reach comparable magnitudes:

$$\text{Free vacuum: } \rho_{\text{CMB}} = +4.87 \times 10^{41} \quad (14)$$

$$\text{Constrained vacuum: } |\rho_{\text{Casimir}}| = \frac{\pi^2}{720 d^4} \quad (15)$$

(Corrected on 30 Sept. 2026: previously denominator 240 – energy density instead of Casimir pressure, as above.)

.4 Cosmic Redshift without Expansion

ξ -Field Energy Loss Mechanism

Thesis

[S] According to this thesis, the observed cosmic redshift arises not from spatial expansion but from energy loss of photons in the omnipresent ξ -field. The wavelength dependence assumed here has been replaced in the current corpus by an achromatic redshift (correction K6 in Doc. 190).

Photons lose energy through interaction with the ξ -field:

$$\frac{dE}{dx} = -\xi \cdot f\left(\frac{E}{E_\xi}\right) \cdot E \quad (16)$$

(Note of 2 Oct. 2026 on notation: here $f(E/E_\xi)$ is an energy-dependent function in the energy loss of the photons, not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)

For the linear case $f\left(\frac{E}{E_\xi}\right) = \frac{E}{E_\xi}$ this yields:

$$\frac{dE}{dx} = -\frac{\xi E^2}{E_\xi} \quad (17)$$

Wavelength-Dependent Redshift

Integration of the energy loss equation leads to wavelength-dependent redshift:

Wavelength-dependent redshift:

$$z(\lambda_0) = \frac{\xi x}{E_\xi} \cdot \lambda_0 \quad (18)$$

where λ_0 is the emitted wavelength and x is the distance traveled.

This formula predicts:

- Shorter wavelength light (UV) shows greater redshift
- Longer wavelength light (radio) shows smaller redshift
- The ratio is $z_1/z_2 = \lambda_1/\lambda_2$

Experiment

Experimental test: Comparison of radio and optical redshifts

- 21cm hydrogen line: $\nu = 1420$ MHz
- Optical H α line: $\nu = 457$ THz
- Predicted ratio: $z_{21\text{cm}}/z_{\text{H}\alpha} = 3.1 \times 10^{-6}$

.5 Structure Formation in the Static ξ -Universe

Continuous Structure Development

In the static T0 universe **[S]**, structure formation occurs continuously without Big Bang constraints:

$$\frac{d\rho}{dt} = -\nabla \cdot (\rho \mathbf{v}) + S_\xi(\rho, T, \xi) \quad (19)$$

where S_ξ is the ξ -field source term for continuous matter/energy transformation.

ξ -Supported Continuous Creation

The ξ -field enables continuous matter/energy transformation:

$$\text{Quantum vacuum} \xrightarrow{\xi} \text{Virtual particles} \quad (20)$$

$$\text{Virtual particles} \xrightarrow{\xi^2} \text{Real particles} \quad (21)$$

$$\text{Real particles} \xrightarrow{\xi^3} \text{Atomic nuclei} \quad (22)$$

$$\text{Atomic nuclei} \xrightarrow{\text{Time}} \text{Stars, galaxies} \quad (23)$$

Energy balance is maintained by:

$$\rho_{\text{total}} = \rho_{\text{matter}} + \rho_{\xi\text{-field}} = \text{constant} \quad (24)$$

.6 Dimensionless ξ -Hierarchy

Energy Scale Ratios

The ξ -relations used here reduce to exact mathematical ratios:

Table 1: Dimensionless ξ -ratios

Ratio	Expression	Value
Temperature	$\frac{T_{\text{CMB}}}{E_{\xi}}$	3.13×10^{-8}
Theory	$\frac{16}{9} \xi^2$	3.16×10^{-8}
Length	$\frac{\ell_{\xi}}{L_{\xi}}$	$\xi^{-1/4}$
Casimir-CMB	$\frac{ \rho_{\text{Casimir}} }{\rho_{\text{CMB}}}$	$\frac{\pi^2 \times 10^4}{960}$

(Corrected on 30 Sept. 2026: in the Casimir-CMB row previously $\pi^2 \times 10^4 / 320$ – with the energy density $\pi^2 \times 10^4 / 960 \approx 102.8$.)

Important

The ξ -relations used here consist of exact mathematical ratios:

- Fractions: $\frac{4}{3}, \frac{3}{4}, \frac{16}{9}$
- Powers of ten: $10^{-4}, 10^3, 10^4$
- Mathematical constants: π^2

No arbitrary decimal numbers occur in these expressions; they are built from ξ and integer structural choices (motivated settings).

.7 Experimental Predictions and Tests

Precision Measurements of Gravitational Constant

T0-theory predicts:

$$G_{\text{T0}} = 6.6745 \times 10^{-11} \text{ m}^3 / (\text{kg} \cdot \text{s}^2) \quad (+0.003 \% \text{ relative to CODATA}) \quad (25)$$

(Corrected on 29 Sept. 2026: previously $6.67430000 \dots \times 10^{-11}$ – these digits are not covered by the calculation; the corpus formula $G = \xi^2 / (4m_e) \cdot C_{\text{conv}} \cdot K_{\text{frak}}$ yields 6.6745×10^{-11} .) (Note of 1 Oct. 2026: C_{conv} is the SI conversion of the single-anchor chain (Doc. 383); its numerical value 7.783×10^{-3} additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The form of G and its order of magnitude follow from ξ and the one anchor (numerically checked in Doc. 383, [K]); with the anchor m_e , G lies at -2.6% , with v at -0.46% (Doc. 383). The agreement to 0.003% arises only with this residual and does not count as an independent precision result (Doc. 190, R141).)

This prediction can be tested by future precision measurements.

Casimir Force Anomalies

Experiment

Prediction: Casimir force anomalies at characteristic ξ -length scale

- Standard Casimir law: $F \propto d^{-4}$
- ξ -field modifications at $d = L_\xi = 10^{-4}$ m
- Measurable deviations through ξ -vacuum coupling

Electromagnetic Resonance

Maximum ξ -field-photon coupling at characteristic frequency:

$$\nu_\xi = \frac{c}{L_\xi} \approx 3 \times 10^{12} \text{ Hz} = 3 \text{ THz} \quad (26)$$

(Corrected on 30 Sept. 2026: previously $\nu_\xi = 1/L_\xi = 10^4 \text{ Hz} = 10 \text{ kHz}$ – the conversion to SI lacked c ; $1/L_\xi = 10^4 \text{ m}^{-1}$ is a wavenumber, the frequency is $c/L_\xi \approx 3 \times 10^{12} \text{ Hz}$.)

Electromagnetic anomalies should occur at this frequency.

.8 Cosmological Consequences

Solution to Cosmological Problems

The following table shows how known problems of standard cosmology appear in the static T0 model [S]. The entries are interpretations in the cosmic sector, which the current corpus leaves aside (P39), not derived solutions:

Table 2: Cosmological problems: Standard vs. T0

Problem	Λ CDM	T0 Interpretation
Horizon problem	Inflation required	Infinite causal connectivity
Flatness problem	Fine-tuning	Geometry stabilizes over infinite time
Monopole problem	Topological defects	Defects dissipate over infinite time
Lithium problem	Nucleosynthesis discrepancy	Nucleosynthesis over unlimited time
Age problem	Objects older than universe	Objects can be arbitrarily old
H_0 tension	9% discrepancy	No H_0 in static universe
Dark energy	69% of energy density	Not required

Parameter Reduction

Parameter count

Parameter count: the Standard Model and Λ CDM together use 25+ parameters, T0-theory the single parameter ξ (plus integer structural choices as motivated settings; SI anchors serve only for conversion).

- Standard model of particle physics: 19+ parameters
- Λ CDM cosmology: 6 parameters
- T0-theory: 1 parameter (ξ)

Arithmetically this corresponds to a 96% reduction. The Λ CDM share concerns the cosmic sector, which the current corpus leaves aside (P39).

Bibliography

- [1] Pascher, Johann (2025). *Vereinfachte Lagrange-Dichte und Zeit-Massen-Dualität in der T0-Theorie*. T0-Theory Project. https://jpascher.github.io/T0-Time-Mass-Duality/2/pdf/129_lagrangian-einfach_De.pdf
- [2] Pascher, Johann (2025). *Simplified Lagrangian Density and Time-Mass Duality in T0-Theory*. T0-Theory Project. https://jpascher.github.io/T0-Time-Mass-Duality/2/pdf/129_lagrangian-einfach_En.pdf
- [3] Pascher, Johann (2025). *T0-Modell: Ein vereinheitlichtes, statisches, zyklisches, dunkle-Materie-freies und dunkle-Energie-freies Universum*. T0-Theory Project. https://jpascher.github.io/T0-Time-Mass-Duality/2/pdf/063_cosmic_De.pdf
- [4] Pascher, Johann (2025). *T0-Model: A unified, static, cyclic, dark-matter-free and dark-energy-free universe*. T0-Theory Project. https://jpascher.github.io/T0-Time-Mass-Duality/2/pdf/063_cosmic_En.pdf
- [5] Pascher, Johann (2025). *Temperatureinheiten in natürlichen Einheiten: T0-Theorie und statisches Universum*. T0-Theory Project. https://jpascher.github.io/T0-Time-Mass-Duality/2/pdf/061_TempEinheitenCMB_De.pdf
- [6] Pascher, Johann (2025). *Temperature Units in Natural Units: T0-Theory and Static Universe*. T0-Theory Project. https://jpascher.github.io/T0-Time-Mass-Duality/2/pdf/061_TempEinheitenCMB_En.pdf
- [7] Pascher, Johann (2025). *Geometric Determination of the Gravitational Constant: From the T0-Model*. T0-Theory Project. https://jpascher.github.io/T0-Time-Mass-Duality/2/pdf/127_gravitationskonstante_En.pdf
- [8] Pascher, Johann (2025). *T0-Theorie: Wellenlängenabhängige Rotverschiebung ohne Distanzannahmen*. T0-Theory Project. https://jpascher.github.io/T0-Time-Mass-Duality/2/pdf/026_T0_Geometrische_Kosmologie_De.pdf
- [9] Heisenberg, W. (1927). *On the intuitive content of quantum theoretical kinematics and mechanics*. Zeitschrift für Physik, 43(3-4), 172–198.
- [10] Planck Collaboration (2020). *Planck 2018 results. VI. Cosmological parameters*. Astronomy & Astrophysics, 641, A6. <https://doi.org/10.1051/0004-6361/201833910>
- [11] CODATA (2018). *The 2018 CODATA Recommended Values of the Fundamental Physical Constants*. National Institute of Standards and Technology. <https://physics.nist.gov/cuu/Constants/>
- [12] Casimir, H. B. G. (1948). *On the attraction between two perfectly conducting plates*. Proceedings of the Royal Netherlands Academy of Arts and Sciences, 51(7), 793–795.

- [13] Muon g-2 Collaboration (2021). *Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm*. Physical Review Letters, 126(14), 141801. <https://doi.org/10.1103/PhysRevLett.126.141801>
- [14] Riess, A. G., et al. (2022). *A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km s⁻¹ Mpc⁻¹ Uncertainty from the Hubble Space Telescope and the SH0ES Team*. The Astrophysical Journal Letters, 934(1), L7. <https://doi.org/10.3847/2041-8213/ac5c5b>
- [15] Naidu, R. P., et al. (2022). *Two Remarkably Luminous Galaxy Candidates at $z \approx 11-13$ Revealed by JWST*. The Astrophysical Journal Letters, 940(1), L14. <https://doi.org/10.3847/2041-8213/ac9b22>
- [16] COBE Collaboration (1992). *Structure in the COBE differential microwave radiometer first-year maps*. The Astrophysical Journal Letters, 396, L1–L5. <https://doi.org/10.1086/186504>
- [17] Sparnaay, M. J. (1958). *Measurements of attractive forces between flat plates*. Physica, 24(6-10), 751–764. [https://doi.org/10.1016/S0031-8914\(58\)80090-7](https://doi.org/10.1016/S0031-8914(58)80090-7)
- [18] Lamoreaux, S. K. (1997). *Demonstration of the Casimir force in the 0.6 to 6 μm range*. Physical Review Letters, 78(1), 5–8. <https://doi.org/10.1103/PhysRevLett.78.5>
- [19] Einstein, A. (1915). *Die Feldgleichungen der Gravitation*. Sitzungsberichte der Preußischen Akademie der Wissenschaften, 844–847.